

Workshopping Participation in Music

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This paper presents a process-based approach to considering workshops as a route to participation in collective creative musical practice. We evoke the notion of the *Music One Participates In*, which focuses on a shift from the listener-as-consumer to participant-actor actively engaged in sound perception and production. We look at the range of different methods that make up the term ‘workshop’, as well as emergent relationships between facilitator and participant in creative do-it-yourself activities to frame our discussion of participatory music practice. We then examine the nature of participation across a range of disciplines from social and cognitive science through human computer interaction to radical and contemporary art, and identify possible contradictions in horizontal utopian organisational models. With this conceptual frame as a backdrop, we present four types of workshop that we have conducted across time at different sites with diverse groups of participants. We apply concepts from the participation literature to analyse our music workshops, and attempt to reconcile the potentially diverging agendas of facilitator and participant to arrive at a process-based view of ‘workshopping’.

1. INTRODUCTION

1.1. Music One Participates In

A founding member of the Creative Commons, Lawrence Lessig, describes a re-emergence of creative acts by ordinary citizens using the term read/write (R/W) culture (Lessig 2008). He argues that, in the twentieth century, with the arrival of recording and broadcasting technologies, ‘We would become practiced in selecting what we wanted to hear, but not practiced in producing stuff for others to hear’ (Lessig 208: 25). In the last decade of the twentieth century, the spread of digital technologies has revived user-centric creativity. People are able to share and remix music on their laptops by using tools similar to those previously reserved for professionals. With these technologies, people do not just ‘read’ culture through consuming but also ‘write’ their own culture by creating and re-creating the media content around them.

In Jo and Tanaka (2009), we draw upon Arnstein’s *Ladder of Citizen Participation* (1969) and evoke Attali’s notion of *composition* (Attali 1985) and Small’s concept of *musicking* (Small 1998) to discuss the participatory nature of a number of musical works. We extend Barthes’ binary division of music into Music One Plays and Music One Listens To

(Barthes 1977a) to propose a *Music One Participates In*. In this article, we develop the notion of participation in music to problematise participation as a complex system of relations where agency and identity enter into play. We look at the acts of organising and taking part in workshops and coin the term ‘workshopping’, following Small’s conversion of *music* as a noun to *musicking* as a process-based act, as a way to understand methods to facilitate musical participation across a range of different stakeholders.

1.2. Workshops

There are a number of different dictionary definitions for the word ‘workshop’. Words typically used to describe and characterise workshops include: *discussion, activity, intensive, skills, interaction, exchange, creative, improvisation*. Workshops may be used in design processes, in ethnographic work to understand users and cultures, or in brainstorming exercises to stimulate innovation. Workshops also exist in academic conferences as a format to facilitate focused and interactive discussion. They are used in industry in the form of focus groups for marketing or as ways to bring about culture change in companies (Nieters and Bollman 2011). In the creative arts, workshops can refer to master classes with mentors and well-known artists or information sessions on techniques and materials for practice. In these contexts, workshops are upstream activities to further research or framing activities to coach artistic practice. Workshops are a central dissemination activity in the creative do-it-yourself (DIY) community for skill and knowledge sharing. From this range of similar but distinct formats, we begin to see that workshop is a word with many meanings, something that takes on multiple forms and formats depending on context.

Methods for the delivery of workshops have been codified in different areas of practice such as design, education and industry. They are most often goal oriented, and typically aim to be effective in realising design tasks, generating learning results and delivering product innovation. In design, Svanæs (Svanæs and Seland 2004) and Westerlund (2007) present step-by-step methods for role-playing, talking and lo-fi prototyping. Nieters (Nieters and Bollman 2011) presents case studies from Yahoo! UX design workshops as strategies to

bring about corporate culture change and build consensus across a group of mixed stakeholders.

Concepts found in education workshops bring up concepts that undo traditional hierarchies and are relevant to themes that will be discussed here. Spikell and Aghevli (1999) look at the effectiveness of workshop methods in mathematics education. They coin useful terms that describe evolving roles of workshop facilitator and participant. In their workshops, they seek to create 'involved and motivated' learners who will allow the workshop leader to be less a 'sage on the stage' and more a 'guide on the side'.

1.3. Do-it-yourself (DIY)

The do-it-yourself (DIY) movement, documented in publications such as *Make* magazine and encountered in its associated festival, Maker Faire, puts an emphasis on shared creativity. The Human–Computer Interaction (HCI) field has taken up DIY to gain insight into technology design and dissemination, with Williams and Irani (2010) identifying in DIY projects approaches based on open creativity rather than task-oriented performance. Buechley defines DIY as an 'array of creative activities in which people use, repurpose and modify' in ways that are 'codified and shared so that others can reproduce, re-interpret or extend them' (Buechley, Rosner, Paulos and Williams 2009: 4824).

Goodman and Rosner (2011) look at the integration of technology into handwork activities and observe three ways in which digital technology is accommodated into physical acts of everyday creativity:

1. *segmenting*, in which activities affecting the object of handwork can take on independent trajectories,
2. *interjecting*, where immediate needs for information or coordination prompt brief switches from one activity to another, and
3. *extending*, in which direct embodied handwork interactions occur as people attend to other stimuli.

These modes suggest the richness with which we can consider participatory practices beyond the simple completion of tasks.

Recently, there are a growing number of workshops that apply DIY methods in teaching music and music technology. *Scrapyard Challenge* is a series of workshops where participants build small computational objects out of discarded electronics parts (Moriwaki and Cohen 2006). Their workshop format works within a number of constraints, including limited time frame, use of found materials, simple input/output and accessible hobbyist electronics. With these components, participants create musical controllers that produce MIDI data using simple but with unorthodox materials such as washboards, baby jumpers and helmets.

Dirty Electronics is a range of workshop activities that connects live bricolage with instrument building and stage performance (Richards 2008). During a Dirty Electronics workshop, participants build sound-generating devices based on prepared kits. After building a device, participants take part in a group performance under the direction of the workshop organiser. In the performances it is common that the participant's own body becomes part of the electronic circuit, using body resistance/capacitance to modulate characteristics of the sound produced.

The study and evaluation of such hands-on musical workshops/performances often overlaps with the documentation of the events. In *Scrapyard Challenges*, the organisers discuss workshops in terms of the instruments/artefacts produced by the participants. In *Getting the Hands Dirty* (Richards 2008), Richards describes a pedagogical role in considering the relationship between the body and electronic instruments in a performance. He also mentions that, for a musician, having built an electronic instrument by her/himself provides the person with a better understanding of the potential use of the device in performance. These views provide insight into how individual and collaborative forms of creativity come together in the context of a workshop/performance and how the workshop process and its output as performance might connect with participants' everyday lives (Allen, Galani and Jo 2009).

Looking at DIY-music workshops, we were interested to find out in what ways they differed from traditional music masterclass-workshops, and conversely the ways in which musical activity could extend established DIY activities. By describing several types of DIY-inspired music technology workshops we have conducted, and looking at the conceptual and theoretical underpinnings motivating them, we seek to identify emerging themes with which we could characterise participatory music activity.

1.4. Participation

In participatory practices, the notion of authority is often challenged and problematised. Content is no longer seen as a static object created by a single author, but instead becomes part of an emergent and sustained relationship between authors (artists, designers, managers) and participants (audiences, users, workers). In order to map notions of participation to music in the era of Lessig's R/W practices, we need first to unwrap the diversity of participation across a range of domains. Similar to the multiple definitions of the word workshop, there are a variety of interpretations of the word 'participation', with each field exploring different aspects of its implementation. In this section, we examine the nature of participation across disciplines from social science through human

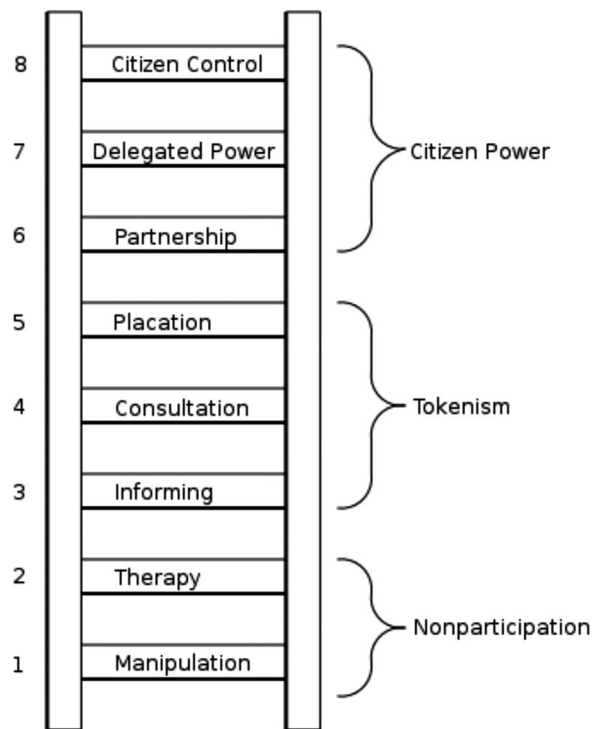


Figure 1. Three levels on the ladder of citizen participation (Arnstein 1969).

computer interaction to contemporary art leading up to today's do-it-yourself (DIY) movement.

1.4.1. The ladder of citizen participation

Participation in social science is an umbrella term to cover decision-making processes in various activities including politics, economics, management and education. Arnstein (1969) defines three broad levels of citizen participation: *non-participation*, *tokenism* and *citizen power* within which are eight rungs (from manipulation to citizen control) of a ladder (Figure 1).

The ladder serves to illustrate gradations of participation in society and depicts potentially deceptive political and economical actions of power holders. Arnstein argues that 'participation without redistribution of power is an empty and frustrating process for the powerless' (1969: 216).

Environmental psychologist Roger Hart extends Arnstein's ladder, using it as a starting point to think about children's participation in social benefit projects (Hart 1992). In collaborative work with UNICEF (the United Nations Children's Fund), he opposes the judgement that 'participation by children is a naive notion for children who simply do not have the decision-making power of adults' (1992: 5) and argues for the importance of the involvement of children, the least listened to members of society, in meaningful projects with adults. To depict circumstances where children's involvement is ambiguous and even subject

to manipulation, Hart proposes four requirements as a basis for a project to be truly considered as participatory:

1. that participants understand the intentions of the project,
2. that they know who makes decisions concerning their involvement and why,
3. that they have meaningful (rather than 'decorative') roles, and
4. that they volunteer for the project once the project has been made clear to them.

Hart's criteria serve to elucidate the real distinction between forms of full participation compared to the token forms of participation midway down Arnstein's ladder. Whether or not we subscribe to political activism, this points out that forms of agency are important to genuine participation. We will make use of these four criteria in describing the kinds of participation in the music workshops we present below.

1.4.2. Participatory design and communities of practice

In HCI, participation is a core part of a family of user-centric methods in qualitative research used to study technology-mediated interaction from workplace design through situated learning to online social networking.

One important tradition of participatory design comes from Scandinavian research practice dating back to the 1960s where researchers discussed approaches to computer systems design with the very people destined to use the system and to engage with them as a critical role in the designing process. Czyzewski et al, describe 'workers' determinability for their work' and attach importance to 'the user's perceptions and feelings about technology'; they view computer systems 'as processes rather than as products' (Czyzewski, Johnson and Roberts 1990: ii). In these approaches, the privilege of the expert is called into question. For the success of a project, it is important to have a partnership between implementers and users.

In the workplace literature, Lave and Wenger view learning as a situated activity (Lave and Wenger 1991) that forms *communities of practice*. They use examples from apprenticeship programmes to describe processes by which newcomers become part of a community centred around shared objectives. They argue that the need to acquire knowledge and skill motivates newcomers to participate fully in the sociocultural practices of a professional community. Preece and Schneiderman extend these notions beyond the workplace by proposing the *Reader-to-Leader Framework* as a way to understand people's participation in online social activities (Preece and Schneiderman 2009). They identify usability and sociability factors at different stages, evolving from *readers* of discussion

boards, blog posts and photos to *contributors* of user-generated content including music, video, or text articles, through *collaborators*, who form tightly connected groups with lively discussion, to *leaders*, who govern activities of an online community by setting and upholding policies, vetting content, or mentoring novices. This recalls the evolution and shift of roles pointed out in maths education workshops by Spikell and Aheveli (1999).

1.4.3. Creativity and participation

Activities in participatory practices often invoke questioning of authority. These practices do not always have clear destinations for their outcomes, such as museums, concert halls or boutiques. Rather, they try to seek a less tangible but palpable personal sense of fulfilment such as making friends, assembling gadgets or acquiring skills. Wang and Kaye (Wang and Kaye 2011) coin the phrase *inventive leisure practices* (ILP) to describe such ‘small-c’ activities – to denote creativity in lower-case, creative acts, those that do not carry artistic pretensions of Big-C Creativity (Boden 1991). Small-c creativity is not goal oriented, but rather process based, where, according to Csikszentmihalyi’s Theory of Flow, people ‘experience the world in novel and original ways’ (Csikszentmihalyi 1997: 25). Wang and Kaye (2011) describe these creative processes as forms of practice that are ‘complex and messy’ and include in them activities such as creative hacking and craft. These activities use appropriation, are decoupled from the market economy, are based on skill and not accreditation, and take place in the absence of a central authority. These characteristics of ILPs are a useful way to distinguish participatory practices from Big-C arts creativity, which is often highly centralised, and is based on recognition linked to level of training.

1.4.4. Paradox in paradise

Art theorists have cast their attention onto participatory art, and, in the process of doing so, have raised issues and concerns about the maturing process of participation and its ultimate robustness in a succession of practices. Bishop (2012) notes a mismatch of social and artistic judgement in participatory art practices. She coined the term ‘delegated performance’ for situations where the artist uses people as a medium to undertake the job on behalf of the artist.

Curator Rudolf Freiling points out a paradox in 1960s participatory art:

By embracing chance, by giving up control, by inviting others to participate in the production of the artwork, by claiming the radical dismantling of traditional systems for evaluating art, these pioneering figures faced a paradox from the very start: how to do away with art

by making art. Art and antiart, and art and life, have always been closely intertwined in this paradox. (Freiling 2008: 34)

American theorist Hal Foster also points out a tendency towards ‘a promiscuity of collaboration’ (Foster 2004: 192). He argues that ‘the discursivity and sociability’ in these works risks being illegible to the viewer. Sometimes the effects of juxtaposition in projects are more chaotic than communicative. He wonders if the illegibility ‘might reintroduce the artist as the principle figure and the primary exegete of the work’ (2004: 194). Such a paradox points out a lingering authoritarian position of the artist in spite of the supposedly horizontal nature of participatory practice. In essence, the question is whether Big-C artists can legitimately create art objects that celebrate small-c creativity.

2. WORKSHOP REPORTS

We report on four types of music workshop conducted in the period 2009–12, each delivered a number of times in a range of contexts to different types of participants.

2.1. How to Turn Your iPhone into a Musical Instrument

2.1.1. Mobile music workshop

‘How to Turn Your iPhone into a Musical Instrument’ introduces an interactive music programming and performance environment to non-specialists, allowing participants to create custom apps on the mobile where sound sampling and digital audio synthesis are controlled by the accelerometer, multi-touch screen and GPS. The workshop takes place over a half-day, with 12–16 participants and 1–2 workshop facilitators. It has been presented nine times in a range of different settings, from schools to arts festivals, with a range of different kinds of participant, from children, to art-school students, to community and social work groups.

A dense set of information is communicated during the workshop, including graphical programming in Pure Data (PD), ad-hoc wireless communications between the participant’s computer and mobile phone, sensor prototyping with mobile sensors controlling a computer-based audio process, and finalisation of a standalone app on the mobile.

The facilitators perform a concert at the end of the workshop that is open to the public. The workshop participants are invited not just to attend the concert, but also to give a demo or perform in the first part of the concert with their newly made instrument.

2.1.2. Lessons learned

We realised that the workshops were functioning as a somewhat surreptitious way to introduce participants



Figure 2. Chiptune Marching Band workshop, Maker Faire Newcastle.

to data-flow programming within PD, an authoring environment which is often considered to be a ‘dark art’ with a steep learning curve and beyond the reach of the non-specialist. Through focusing on a specific, achievable outcome – the creation of an accelerometer-controlled synthesiser – and not dwelling on the complexity of the tools involved in creating nor the theories of signal processing, we found that many workshop participants very quickly grasped the basics of PD and overcame elements of the learning curve which has been known to put people off using such software. Participants were able to create complex ‘patches’ in a very short time. At the Lisbon workshop, held at IPA on 27 and 28 October 2011, one participant who was experienced as an improviser and programmer, but with no previous experience using Pure Data or performing with an iPhone, was able to develop a patch within a 6-hour workshop that he then performed with in concert that evening. The Tyneside Factory workshop (28 November 2011) was a mere 80 minutes long, yet some participants – 15-year-olds with no experience in programming – were able to not only implement the workshop synth, but also develop and extend it.

Only a few participants took up the challenge and invitation to perform, instead allowing the workshop facilitators to play the concert. The workshop participants hesitated to transfer their small-c experience to a Big-C performance. At the same time, according to participants, it was useful for them to see how what they had just learned could be used in a professional, virtuosic presentation. The concert thus served as an effective foil to the workshop. They were complementary, one enhancing an empathic understanding of the other: the performance showing the potential application of the technologies presented in the workshop, and the workshop making the smoke and mirrors behind the performance less opaque.

2.2. Chiptune Marching Band

2.2.1. DIY musical instruments

‘Chiptune Marching Band’ (CTMB) is a workshop on DIY practice and participatory music where participants make a very simple, personalised musical instrument based on a self-powered sound-producing

analogue circuit, and give a street performance with the other participants (Jo, Allen and Galani 2009). The workshop broaches the shift of scale and focus from the individual to the community – where people who once passively consumed music become involved in its creation. Interrelationships between music, instruments and environments are explored through performance.

The CTMB has taken place in seven different situations, including grassroots artist-run centres (NK Berlin, Space London), large DIY events (Maker Faire Newcastle), art festivals (Bent Festival New York, Pixelache Helsinki) and academic conferences (ACM Creativity and Cognition Berkeley, SIGGRAPH ASIA Yokohama). The workshop lasts 2–3 hours with three sections: Introduction to Concepts, Circuit Building and Instrument Personalisation. The introduction presents concepts of localised resource communities, alternative power, basic electronics – op amps and sensors – and collective musical performance strategies.

During circuit building, participants build an analogue sound producing circuit on a breadboard following step-by-step instructions and power it with a hand-crank generator. They choose a capacitor to set the tonal bandwidth, and select from one of three types of sensor (linear fader, rotary potentiometer and photo-resistor) to control the frequency output. The circuit and sensor are housed in a cardboard tube that can be personalised with colourful tape, glue, paint, craft paper and stickers. Through the sessions, the organisers answer questions and provide assistance.

The self-powered nature of the instruments allows them to be played untethered from the power grid. The group discusses how a chiptune instrument ‘marching band’ might perform, and then go outdoors. The performance takes place in public space (street or public square) as a happening and spectacle (Figure 2).

2.2.2. Lessons learned

The CTMB had a diverse range of participants across and within sessions, including DIY enthusiasts, teachers, high school and art school students, amateur musicians, composers and media artists. The background of each participant had an effect on which stage of the workshop presented a challenge or stumbling



Figure 3. Aeolian harp workshop.

block – be it clearly challenging steps such as circuit building, to less obvious hurdles such as gluing, personalising or performing.

Marching tactics in the CTMB performance varied from group to group and were the most challenging phase of the workshop for the facilitators. The performance, in the spirit of the workshop, was open ended. However, with no performance plan, it often devolved into a sociable walk by a group of people with strange cardboard tubes, and did not have the impact of a Happening to passers by. While social performance is embedded in everyday life, musical performance does not come naturally or automatically, no matter how empowering the act of instrument building. Ensemble musical performance is typically facilitated by conductors and bandleaders. It was difficult, once the workshop facilitators had established themselves during the workshop as, to use Spikell and Aghevli's (1999) terms, a 'guide on the side' to impose themselves as a 'sage on the stage'.

2.3. Aeolian harp

2.3.1. Generative music

In this workshop, we invite participants to build and perform a historical instrument, the Aeolian harp. The workshop allows participants to learn the history and the principle of the instrument, and to listen to the streams of wind with the string vibrations. The workshop has taken place twice – once in an art festival (Koganecho Bazaar) and then at a local artist-run space in Tokyo (403 forbidden) over a half-day for each, with roughly ten participants facilitated by the Generative Music Workshop (Kaneko and Jo 2011) and the instrument builder/researcher Koichiro Sugiyama (Sugiyama 2010).

The workshop consists of three sessions: Lecture, Instrument Building and Performance. The lecture starts with a historical review of the instrument from its origins in Greek antiquity through its peak in the eighteenth century. Participants then learn its

physical structure with basic acoustic theory. During instrument building, participants glue wood boards, blocks and bridges, pound turning pins, and pitch strings over them following a detailed drawing by the facilitators. In the performance, participants take their own instrument outside and explore the field to find a stream of wind to produce sounds (Figure 3).

2.3.2. Lessons learned

In the building process, participants made their own instrument according to the straightforward structure of a prepared kit. Because of the deviation from general musical instruments (e.g. tuning strings to the same pitch to make them resonate each other), participants' pre-existing musical knowledge had little effect in their final outcome.

During the performance, we realised that the structure of the instrument to leave the production of sounds to the wind stimulated participants to perform in a public space. Participants explored the potential of the instrument in diverse ways such as walking in a line while putting the instrument on their heads, running with the instrument, or staying at a corner to wait for a gust of wind. Participants did not baulk at letting the wind resonate the instrument, rather than 'playing' it by blowing on it themselves.

2.4. Social Inclusion through the Digital Economy

2.4.1. Interactive music and social benefit

We conducted a series of participatory music workshops as part of a major UK research project, Social Inclusion through the Digital Economy (SiDE). The project looked at ways in which access to innovative digital technologies could have life transforming effects on young, old and disabled people who were at risk of being marginalised in society. SiDE was a large multidisciplinary project covering areas of health, transport, home and business. The music workshops took place in a part of the consortium dealing with the creative industries.

We worked with young people, 14–22 years old, in multiple groups of ten to fifteen. These young people had no formal musical training and were often on the margins of society (Tanaka, Gaye and Richardson 2010). They were considered the generation of ‘digital natives’, comfortable with mobile and digital technology by having grown up in an already digital society. The top-level premise was that by introducing new ideas to them through media with which they were familiar we would provide gateways to access and acceptance of novel, creative and musical ideas. By demonstrating how mobile technologies could allow users to be musically expressive, we hoped that one outcome of our work would be to undo assumptions that, ‘technology comes from somewhere else’ or that ‘art is done by others’, and show that the very technologies of media consumption could become instruments of personal expression and everyday creativity.

2.4.2. *Lessons learned*

We conducted music technology workshops in partnership with existing community programmes such as the Regional Youth Work Unit (RYWU), Generator Music and education programmes of cultural venues such as the Sage Gateshead. In these workshops, our proposition was to provide interactive music technologies that might not otherwise be available to these local and grassroots organisations.

Interestingly, these technologies were not always seized upon for their musical potential. A completely participatory design approach in the RYWU workshop included the young people in the design of the actual activity we would do with them. After an introduction to interactive music technologies, the participants re-framed the workshop activity to produce a video game like interactive information resource on education pathways. While at first this seemed to shift the focus away from the original musical intentions of the workshop, it underlined the core benefit of the workshop approach as not the creative activity itself (Big-C), but an awareness of the participant’s own agency in creative pursuits (small-c). In the case of interactive technology, a key epiphany for the young participants coming from an economically depressed post-industrial area of the UK was the realisation that advanced technology happened in their own backyard, and not just in far away places such as Japan or Silicon Valley. Their desire to seize this opportunity to take it away from a leisure-oriented one (music) towards a career-development focus (education) demonstrates a sense of empowerment.

The partnerships with community organisations meant that our contact with participants was brokered by gatekeepers. This exposed an important and

complex dynamic of multiple, tiered beneficiaries where there were at times conflicting interests across the facilitators (us), gatekeepers and participants.

3. DISCUSSION

We have highlighted common themes as well as differing perspectives on participation across a range of disciplines. We draw upon dynamics of the DIY movement to emphasise process and activities amongst participants rather than the completion of a task or the production of art objects. From the ladder of participation, we have invoked a degree of involvement in musical practice along a spectrum of participation. From the discussion of participatory design and communities of practice, we have gained a notion of a shift of roles and the evolving processes involved in participation. From these points of view, we take the emphasis away from making objects and instead focus on building events with participants. Through our review of the literature, we have noticed one clear, shared argument for the absolute privilege of the expert’. Freiling proposes a paradox of ‘how to do away with art by making art’ (2008: 34) and Foster ironically notes a concern for ‘the artist as principle figure’ in participatory art (2004: 194). In the following sections, we examine our workshops from these emergent points of view.

3.1. *Rather than completion of a task*

Our workshops are cultural, non-workplace-based and incorporate creative practice. Although they may involve the completion of a task such as the building of a synthesiser, these are not the primary goals of the workshops. Each workshop has a physical component, be it the CTMB circuit or the Aeolian harp. Engagement with tangible elements can be considered ‘handwork’, in Goodman and Rosner’s (2011) terms. Their three ways of accommodation of technology into the everyday, *segmenting*, *interjecting* and *extending*, can be applied here as a way to understand how participants in DIY music workshops deal with the introduction of new concepts, tools and exercises. Our workshop formats fit each of these modes – CTMB and Aeolian harp *segmenting* conceptual discussion, technical building, personalisation and performance. Through the workshop, participants were ‘dividing their time between different genres of activity’: learning background theory and history, making instruments along to instructions, fabricating outcomes in their own way, and playing with them. These segmentations could accommodate the engagement with objects in ‘independent trajectories’ (Goodman and Rosner 2011: 2265).

The SiDE workshops *interject* mobile music technologies into regional social engagement. Even to the

participants who were considered the generation of 'digital natives', the provided interactive music technologies were hardly noticeable to their local and grassroots organisations. The interjected technologies, and the local researchers who presented them, made the participants aware, especially those coming from an economically depressed post-industrial area, that advanced technology development was happening in their own backyard. The interjections could not only involve having fun with musical activities but also fulfilled participants' desire to connect the technologies for their prospective careers.

The iPhone Musical Instrument workshop *extends* a music technology workshop with a concert. Even though the core of the workshop is to introduce participants to interactive music programming, the embedded concert provides opportunities for them to put their new-found knowledge into action, not just attending it but also having a demo or a performance with their newly made instrument. Because of the differences between participants, only a few took up the proposition; however, such an extension suggests 'possibilities for improvisational creativity' such as the case of a musician with no previous technical experience using the programming environment to extend his or her own traditional instrument within a 6-hour workshop.

3.2. Spectrum of participation

In our workshops, we do not require nor seek artistic accomplishment based on virtuosic skill or scholarly knowledge. Instead, we try to explore small-c musicking in our practices with participants who voluntarily take part in the workshops. We look at dynamics of participation in our musical practices where the public enters into the process of co-creation to shift from being a listener-as-consumer, to actor, actively involved in sound production. Rather than compare politicians and citizens, or adults and children, as Arnstein (1969) and Hart (1992) do, we use the ladder of participation with three levels: *non-participation*, *tokenism* and *citizen power* to examine the relationships and dynamics between authors and participants in the musical practices. We do not use the word participation with the assumption it is a 'good thing' (Graham and Cook 2010: 112), but examine its possibilities and limitations along the ladder. Instead of explicitly defining each rung as Arnstein and Hart do, we treat the ladder as continuous spectrum of participation.

We apply Hart's four criteria for meaningful participation to arrive at the following forms of agency that participants along the spectrum of participation exhibit: *understanding intentions*, *knowing who made decisions*, *having meaningful roles* and *consensual participation*.

In the SiDE workshops, deploying music technology in social-work situations pointed out the fact that there is not a single type of workshop participant. In addition to the workshop organisers and the young people, there were the social workers who, as facilitators, accompanied the young people through a range of activities beyond the workshop. These facilitators are often quick to grasp the potential of technology and imagine the enhancement of activities they deliver, while the young participants may lack the confidence to express themselves or think through a hunch or question they may have, but creating an environment conducive to non-hierarchical discussion can bring out unexpected results. We can use the metaphor of the ladder of participation to think not just about different levels of participation, but of the different types of stakeholder who come together in the workshop.

This can result in a return of hierarchy despite all good intentions. In the iPhone musical instrument workshops, the expertise and virtuosity of the facilitator intimidated participants and actually hindered participation by novice participants in the final performances, which was supposed to be open to all. In CTMB, we held non-hierarchical discussions amongst facilitator and participants to decide on marching tactics and performance scenography. However, the outcome was more often than not unclear and resulted in performance that was chaotic and 'promiscuous' in Foster's (2004) terms. Participants in the Aeolian harp workshop had a well-structured workshop with a lecture and an instrument-building, and a performance by experts brought few unexpected results.

3.3. Shift of roles

In our workshops, we try to include participants in a process of building a (temporary) community of practice, drawing upon Czyzewski's work environments (Czyzewski et al. 1990) and applying it in the small-c creative workshop context. The process moves from mutual conversation with self-introduction and chatting in a casual manner, to a transfer of knowledge and skills, a programming/building/making stage, and a final presentation which includes performance. We view the workshops 'as processes', while trying to maintain participants' sense of 'determinability for their work' (Czyzewski et al. 1990: ii).

One challenge in conducting these workshops is the diverse range of previous experience amongst the participants. For example, in the case of CTMB, we always had some participants who were already familiar with technical aspects (e.g. circuit building), while others in the same workshop had no prior electronics experience. We observed the emergence of mutual teaching amongst participants, a kind of self-organising peer-learning dynamic based on prior

experience. Some even made suggestions for circuit customisation to the workshop organiser. The situations typically depict the exchange and contribution of knowledge in apprenticeship described by Lave and Wenger (1991) and in sociocultural activity described by Preece and Schneiderman (2009).

3.4. Away from the paradox

The paradox of the artist's own place in participatory art casts a concern on our practices in terms of its genuineness. Can we consider our workshops art projects at all? As Bishop noted, from an artistic point of view, 'art and the social are not to be reconciled' (Bishop 2012: 278). It is problematic to think about our workshop practices from a Big-C, goal-oriented point of view – there were the aspects that were troublesome, such as the hesitation in performing in the iPhone musical instrument workshop, and the hurdles to organising a successful march in CTMB. On the other hand, by thinking of these workshop activities from a process-based, small-c point of view, we see how creative ideas can be triggered off in participants. In the Aeolian harp event, we could observe an exceptional style of performance (e.g. walking in a line, running with the instrument, staying in a corner) because of the nature of the instrument actuated by wind. In SiDE, there was appropriation by participants who went as far as re-framing and almost hijacking original workshop objectives introducing interactive music technologies to producing a video-game-like information resource.

4. CONCLUSION

We have proposed a process-based approach to considering workshops as a delivery instrument for emergent forms of music. The goal is not to make musicians out of workshop participants, but to use the activity of sound exploration as a vehicle for engagement and possible conduits to unlock forms of creativity embedded in the everyday. In order to frame our discussion, we explored the nature of participation in different fields, and applied these concepts in our discussion of a series of sound and music workshops with a diverse range of participants. The insights gained not only allow us to think of workshops as a methodology but also enable us to understand the transfer of authority in existing music practices. Eco, in describing open works, notes a shift of initiative from the composer to the individual performer (Eco 1959). He notes a difference between works that a composer arranges in a closed, well-defined manner before presentation, and works that are brought to their conclusion at the moment of their realisation with multiple formal possibilities.

In the different relationships and forms of sharing that take place in music, people subjectively encounter music through listening. Our workshops take up Small's (1998) activation of music as the act of musicking through a process of *workshopping* to encourage participants not just to listen but also to proactively engage with the creation of sound. In these projects, each person is a listener of others, and a performer to others. The traditional boundaries separating the artist, composer or writer from spectator becomes blurred. The creator of sound does not prescribe the resulting music, nor explicitly determine meaning to be transmitted. Instead, participants situate themselves in the environment and trust the actions of other participants. This transfer of control and intention has the potential to bring about unpredictable individual experiences for each respective participant. Meanwhile, the organising artist's role in these contexts is far from obviated. Instead of disappearing completely, the artist becomes author of the situation, a form of 'meta-designer' (Giaccardi 2005), intervening in perspectives of sound-making practice by sculpting levels of participation. The resulting representation is a collaborative one that is empowering for both author and participant, but is not a flat hierarchy, and can preserve defined, albeit non-traditional roles.

In the first half of the twentieth century, Benjamin described the decay of aura in art with the advent of mechanical production (Benjamin 1936). In the latter half of the century, Barthes pointed out the death of the author in conjunction with the birth of the reader (Barthes 1977b). In the early part of the present century, the notion of co-creation is gaining currency. We hope that this presentation of workshopping as a route to participation in music might establish a basis of further discussion around ideas of collective creative musical practice.

REFERENCES

- Allen, J., Galani, A. and Jo, K. 2009. An Ecology of Practice: Chiptune Marching Band, *Proceedings of ACM Creativity and Cognition Conference 2009*, New York: ACM Press, 347–8.
- Arnstein, S.R. 1969. A Ladder of Citizen Participation. *Journal of the American Institute of Planners* 35(4): 216–24.
- Attali, J. 1985. *Noise: The Political Economy of Music*, trans. Brian Massumi. Minneapolis: University of Minnesota Press.
- Barthes, R. 1977a. Musica Practica. In *Image-Music-Text*. New York: Hill and Wang, 149–54.
- Barthes, R. 1977b. The Death of the Author. In *Image-Music-Text*. New York: Hill and Wang, 142–8.
- Benjamin, W. 1936. The Work of Art in the Age of Mechanical Reproduction. In *Illuminations: Essays and Reflections*, ed. H. Arendt, trans. H. Zohn. New York: Schocken, 1978, 217–53.

- Bishop, C. 2012. *Artificial Hells: Participatory Art and the Politics of Spectatorship*. London: Verso.
- Boden, M. 1991. *The Creative Mind: Myths & Mechanisms*. New York: Basic Books.
- Buechley, L., Rosner, D., Paulos, E. and Williams, A. 2009. DIY and CHI: Methods, Communities, and Values of Reuse and Customization. In *CHI'09 Extended Abstracts on Human Factors in Computing Systems*. New York: ACM Press, 4823–6.
- Csikszentmihalyi, M. 1997. *Creativity: Flow and the Psychology of Discovery and Invention*. New York: Harper Perennial.
- Czyzewski, P., Johnson, J. and Roberts, E. 1990. Introduction: Purpose of PDC-90. In *Proceedings of the Conference on Participatory Design, PDC'90*. Palo Alto, CA: Computer Professionals for Social Responsibility, ii–iii.
- Eco, U. 1959. The Poetics of the Open Work. In C. Cox (ed.) *Audio Culture: Readings in Modern Music*. New York: Continuum, 1999, 167–75.
- Foster, H. 2004. Chat Rooms. In C. Bishop (ed.) *Participation*. London and Cambridge, MA: Whitechapel Gallery/The MIT Press, 2006, 190–5.
- Frieling, R. 2008. Toward Participation in Art. In R. Frieling (ed.) *The Art of Participation: 1950 to Now*. London: Thames & Hudson, 32–49.
- Giaccardi, E. 2005. Metadesign as an Emergent Design Culture. *Leonardo Music Journal* 38.4: 342–9.
- Goodman, E. and Rosner, D.K. 2011. From Garments to Gardens: Negotiating Material Relationships Online and 'By Hand'. *Proceedings of ACM CHI 2011*. New York: ACM Press, 2257–66.
- Hart, R.A. 1992. *Children's Participation: From Tokenism to Citizenship*. Innocenti Essays No. 4. Florence: UNICEF International Child Development Centre.
- Jo, K. and Tanaka, A. 2009. The Music One Participates In. In F. Schroeder (ed.) *Performing Technology: User Content and the New Digital Media*. Newcastle upon Tyne: Cambridge Scholars Publishing, 34–50.
- Jo, K., Allen, J. and Galani, A. 2009. Chiptune Marching Band: A Public Workshop and Performance. *Proceedings of ACM Creativity and Cognition Conference 2009*. New York: ACM Press, 481–2.
- Kaneko, T. and Jo, K. 2011. Generative Music Workshop. In *Proceedings of Culture and Computing 2011*, 179–80.
- Lave, J. and Wenger, E. 1991. *Situated Learning: Legitimate Peripheral Participation*. Cambridge: Cambridge University Press.
- Lessig, L. 2008. *REMIX: Making Art and Commerce Thrive in the Hybrid Economy*. New York: Penguin.
- Moriwaki, K. and Brucker-Cohen, J. 2006. Lessons from the Scrapyard: Creative Uses of Found Materials within a Workshop Setting. *AI & Society* 20(4): 506–25.
- Nieters, J. and Bollman, E. 2011. Leading Change with Collaborative Design Workshops. *Ext. Abs. CHI'11 Extended Abstracts on Human Factors in Computing Systems*. New York: ACM Press, 729–39.
- Preece, J. and Schneiderman, B. 2009. The Reader-to-Leader Framework: Motivating Technology-Mediated Social Participation. *AIS Transactions on Human-Computer Interaction* 1(1): 13–32.
- Richards, J. 2008. Getting the Hands Dirty. *Leonardo Music Journal* 18: 25–31.
- Small, C. 1998. *Musicking: The Meanings of Performing and Listening*. Hanover, NH: University Press of New England.
- Spikell, M. and Aghevli, B. 1999. The Workshop Method of Teaching: An Example from the Discipline of Mathematics Education. In *Inventio* 2(1).
- Sugiyama, Kouichiro. 2010. *An Empirical Study on Aeolian Harp as a 'Listening' Apparatus in Urban Environment*. PhD thesis, Kyushu University.
- Svanæs, D. and Seland, G. 2004. Putting the Users Center Stage: Role Playing and Low-Fi Prototyping Enable End Users to Design Mobile Systems. *Proceedings of ACM CHI 2004*. New York: ACM Press, 479–86.
- Tanaka, A., Gaye, L. and Richardson, R. 2010. Co-Production and Co-Creation: Creative Practice in Social Inclusion. In R. Nakatsu, N. Tosa, F. Naghdy, K. Wong and P. Codognet (eds.) *Cultural Computing*. Berlin: Springer, 169–78.
- Wang, T. and Kaye, J. 2011. Inventive Leisure Practices: Understanding Hacking Communities as Sites of Sharing and Innovation. *CHI'11 Extended Abstracts on Human Factors in Computing Systems*. New York: ACM Press, 263–72.
- Westerlund, B. 2007. A Workshop Method that Involves Users Talking, Doing and Making. In *Proceedings of the International Conference on Human-Machine Interaction, Human07*, IEEE (March).
- Williams, A.M. and Irani, L. 2010. There's Methodology in the Madness: Toward Critical HCI Ethnography. *Proceedings CHI '10 Extended Abstracts*: 2725–34.